METRIC

MILITARY SPECIFICATION

PIN, STRAIGHT, HEADLESS (DOWEL), GENERAL SPECIFICATION FOR, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers standard size and oversize metric headless dowel pins, commonly referred to as machine dowel pins.

1.2 Classification. The headless dowel pin shall be of the following classes as specified:

Class 1 - Dowel standard size metric (DOD-P-63478/1)
Class 2 - Dowel oversize metric (DOD-P-63478/2)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electrodeposited).
PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging and Packing for Shipment and Storage of.

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DOD-P-16232 - Phosphate Coating, Heavy, Manganese or Zinc Base (For Ferrous Metals).
DOD-P-63478/1 - Pin, Straight, Headless, (Dowel), Standard Size, Metric.
DOD-P-63478/2 - Pin, Straight, Headless, (Dowel), Oversize, Metric.
MIL-C-13924 - Coating, Oxide, Black for Ferrous Metals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TST-S, Dover, NJ 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-66 - Steel: Chemical Composition and Hardenability.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)/AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ANSI/ASTM E18-79 Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1-1978 - Surface Texture

(Applications for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and specification sheet, the latter shall govern.

3.2 Materials. Recycled and reclaimed materials (see 3.1) shall be used to the maximum extent practicable.

3.2.1 Carbon or alloy steel. Carbon or alloy steel dowel pins shall be in accordance with FED-STD-66, except that the maximum sulphur and phosphorous of the composition shall not exceed 0.05 and 0.04 percent respectively.

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3.3 Dimensions. Dimensions and tolerances for all types of dowel pins shall conform to the dimensional requirements of the applicable specification sheet and shall apply after protective finishes.

3.4 Performance characteristics. The dowel pins shall meet the following performance requirements when subjected to the applicable test specified in 4.8.

3.4.1 Double shear strength. Dowel pins should be capable of withstanding the minimum double shear strengths specified in table I.

TABLE I. Double shear strength.

Nominal pin dia. (mm)	kN (kilonewtons) min
1.5	3
2	6
2.5	9
3	13
4	22
5	35
6	51
8	90
10	141
12	203
14	276
16	360
20	563
25	880

3.5 Finish. The protective finish of the dowel pins when specified in the applicable specification sheet shall be as follows (see 4.8.3).

3.5.1 Carbon and alloy steel dowel pins. When phosphate coating or cadmium plating is specified for carbon or alloy steel dowel pins (see 3.1), they shall conform to 3.5.1.1 or 3.5.1.2 as applicable. When uncoated dowel pins are specified in the contract or order, black-oxide coated dowel pins per MIL-C-13924, class 1 may be furnished at the option of the contractor.

3.5.1.1 Phosphate coating. Carbon and alloy steel dowel pins shall be phosphate coated in accordance with DOD-P-16232, type Z, class 2. Phosphate coating shall be 5 to 15 µm (micrometers) thick.

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3.5.1.2 Cadmium plate. Carbon and alloy steel dowel pins shall be cadmium plated in accordance with QQ-P-416, type II, class 2. Plating thickness shall be 7.6 µm (micrometers).

3.6 Hardness.

3.6.1 Carbon or alloy steel dowel pins. Carbon or alloy steel dowel pins shall have a core hardness of 40HRC to 58HRC or equivalent, and a case hardness of 58HRC to 63HRC or equivalent. However, the core hardness of the dowel pin shall not in any instance exceed the case hardness. Minimum penetration for the case hardness shall be 0.4 mm minimum thru 10 mm diameter and 0.75 mm minimum over 10 mm diameter. These requirements shall be met when tested in accordance with 4.8.4.

3.7 Straightness. Dowel pins shall be straight over the effective length within an accumulative total of 0.013 per 25 mm of length for nominal lengths, up to and including 100, and within 0.05 mm total for all nominal lengths over 100 mm (see 4.8.5).

3.8 Surface roughness. Prior to coating, the surface roughness of the cylindrical surface (diameter) of the dowel pins shall not exceed 0.4 µm and all other surfaces shall not exceed 3.2 µm when defined in accordance with ANSI B46.1.

3.9 Workmanship. The workmanship shall be uniform in quality and free of irregularities or detrimental defects. The surfaces shall not contain any foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Material inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.8.1, 4.8.2, 4.8.3, 4.8.5 and 4.8.6.

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4.4 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials used in fabricating the dowel pins are in accordance with the applicable paragraph listed in 3.2.1.

4.5 Quality conformance inspection. Quality conformance inspection shall be as specified in table II. Each dowel pin which does not pass any of the tests specified in paragraph 4.8 shall be defective.

4.6 Quality conformance inspection sampling.

4.6.1 Inspection lot. An inspection lot shall consist of all dowel pins covered by a single specification sheet produced under essentially the same conditions, and offered for inspection at any one time.

TABLE II. Quality performance inspection.

Inspection	Requirement paragraph	Test method paragraph
Dimensions	3.3	4.8.1
Double shear strength	3.4.1	4.8.2
Finish	3.5.1.1, 3.5.1.2	4.8.3
Hardness	3.6.1	4.8.4

4.6.2 Sampling for visual and dimensions examination. A random sample of dowel pins shall be taken from each lot in accordance with MIL-STD-105 inspection level I. The acceptable quality level (AQL) shall be indicated in table III. The AQL shall apply to each individual defect, not to a group of defects.

4.6.3 Sampling for shear test. Sampling for shear test of dowel pins shall be in accordance with level S-1 of MIL-STD-105. The AQL shall be 1.5 percent defective.

4.6.4 Sampling for finish test. Sampling for protective finish test of dowel pins shall be in accordance with the applicable specifications referenced in 3.5.1.1 and 3.5.1.2.

4.6.5 Sampling for hardness test. Sampling for hardness test of dowel pins shall be in accordance with level S-1 of MIL-STD-105. The AQL shall be 1.5 percent defective.

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TABLE III. Classification of defects.

Category	Defect	Inspection method
Critical	None defined	
Major 101 102	AQL = 1.0 percent defects Diameter Straightness	 SIE 1/ SIE 1/
Minor 201 202	AQL = 2.5 percent defects Protective finish Workmanship	 Visual Visual

1/ Standard inspection equipment.

4.7 Packaging inspection. The sampling and inspection of the preservation, packaging, packing and container marking shall be in accordance with the requirement of PPP-H-1581.

4.8 Methods of inspection.

4.8.1 Visual and dimensional examination. Each dowel pin taken as specified in 4.6.2 shall be examined to verify conformance with this specification. Examination shall be conducted in accordance with table III.

4.8.2 Double shear strength test. Each dowel pin taken as specified in 4.6.3 shall be tested for double shear strength in accordance with 4.8.2.2.

4.8.2.1 Shear test fixture. The shear test shall be made by means of a suitable fixture which meets the following requirements. The dowel pin shall be placed in a hole that passes through a shear block and a plunger within the block. The maximum clearance between the shearing planes of the block and the plunger shall be 0.127 mm. The block and plunger shall be so constructed that the shear planes are normal to the longitudinal axis of the dowel pin being tested. The block and plunger shall be made of hardened steel or shall have steel inserts with a minimum shearing-surface hardness of 65HRC. The clearance or interference between the dowel pin and pinhole of the block and plunger shall be within the following limits.

Nominal pin dia. (mm)	Max clearance (mm)	Max interference (mm)
1.5 to 5	.0075	.0050
6 to 10	.0100	.0075
12 to 16	.0125	.0075
20 to 25	.0150	.0100

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The dowel pin to be tested shall be assembled to the block and plunger with the two ends of the dowel pin at least one dowel pin diameter away from the shear planes. A load equal to the minimum double shear value specified in table I for the dowel pin size being tested, shall be applied to the plunger. If fracture occurs in the dowel pin under the applied shear load, dowel pin shall be considered defective. Dowel pins too short to be tested in double shear shall be tested by applying the test to two dowel pins simultaneously in single shear.

4.8.2.2 Double shear test method. The dowel pins shall be tested by the double shear method and shall have the minimum double shear strength as shown in table I.

4.8.3 Finish test. The carbon or alloy steel dowel pins shall conform to the finish test of QQ-P-416 or DOD-P-16232.

4.8.4 Hardness test. The test procedure shall be in accordance with ASTM E18, except that the core hardness of each specimen shall be measured on a transverse section of a point one-eighth of the nominal diameter from the axis of the dowel pin. The case hardness shall be determined with a Rockwell Superficial hardness tester in accordance with ASTM E18 using a 15 kilogram load.

4.8.5 Straightness test. The straightness of the dowel pins shall be determined by using one of the following test methods.

4.8.5.1 Ring gage method. The straightness over the length of the dowel pins shall be such that the dowel pins will pass freely through a ring gage of length as tabulated below for the respective dowel pin lengths. The maximum inside diameter of the gage shall be greater than the maximum diameter specified for the pins on the applicable specification sheet by the amount of the straightness tolerance.

Dowel pin length mm	Gage length mm	
	max	min
to and incl 25	25.15	24.85
over 25 to 50	50.15	49.85
over 50	75.15	74.85

4.8.5.2 Indicator gage method. The dowel pin shall be supported at each end in V- Blocks, rotated beneath a dial indicator, measuring FIM (Full Indicator Movement). The maximum FIM allowable will be twice the straightness tolerance as specified in 3.7.

4.8.6 Surface roughness test method. Surface roughness shall be checked by using surface texture equipment capable of measuring the requirement specified in 3.8.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging, packing and container marking shall be in accordance with PPP-H-1581 (see 6.2).

6. NOTES

6.1 Intended use. Dowel pins covered by this specification are intended for use in machine applications, employing light drives for torque transmission, positioning parts with respect to one another, and for fastening together two or more parts of an assembly.

6.2 Ordering data. Procurement documents should specify the following:

a. Title number and date of this specification and applicable specification sheet.
b. Applicable military specification sheet part number(s).
c. Applicable levels of packaging protection (see 5.1).

6.3 Definitions.

6.3.1 Metric terms and definitions. Metric terms used in this specification are defined in ASTM E380, standard for metric practice.

Custodians:
Army - AR
Air Force - 99

Review activities:
Army - AT
Air Force - 11
DLA - IS
NSA - NS

User activities:
Army - ME

Preparing activity:
Army - AR

Agent:
DLA-IS

(Project 5315-0317)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

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DOCUMENT IDENTIFIER AND TITLE

PIN, STRAIGHT, HEADLESS (DOWEL), GEN. SPEC. FOR METRIC DOD-P-63478

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

[ ] YES [ ] NO *(If "Yes", in what way?)*

4 REMARKS

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DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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NOTICE
OF VALIDATION

METRIC
DOD-P-63478
NOTICE 1
14 April 1987

MILITARY SPECIFICATION

PIN, STRAIGHT, HEADLESS (DOWEL), GENERAL SPECIFICATION FOR, METRIC

DOD-P-63478, dated 29 May 1981, has been reviewed and determined to be valid for use in acquisition.

Custodians:
Army - AR
Air Force - 99

Review activities:
Army - AT
Air Force -11
DLA-IS
NSA-NS

User activities:
Army - ME

Preparing activity:
Army - AR

AMSC N/A FSC 5315

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METRIC
DOD-P-63478
AMENDMENT 1
16 February 1990

MILITARY SPECIFICATION
PIN, STRAIGHT, HEADLESS (DOWEL), GENERAL SPECIFICATION FOR, METRIC

This amendment forms a part of DOD-P-63478, dated 29 MAY 1981, and is approved for use by all Departments and Agencies of the Department of Defense.

Page 3

3.3: Add the following:

"The dimensions of phosphate coated pins shall apply prior to phosphate coating."

Custodians:
Army - AR
Air Force - 99

Review activities:
Army - AT
Air Force - 11
DLA - IS
NSA - NS

User activities:
Army - ME

Preparing activity:
Army - AR

(Project 5315-0457)

AMSC N/A
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